TWO-SIDED LIMITS

The existence condition

A two-sided limit exists if and only if both one-sided limits exist and are equal.

EXISTENCE THEOREM

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = L \text{ and } \lim_{x \to a^+} f(x) = L$$

HOLDS WHEN the left-hand limit exists, the right-hand limit exists, and the two values are identical

✓ The limit exists

Left-hand and right-hand limits both exist and are equal

All three conditions hold; the common value is L.

BOUNDARY

✗ The limit does not exist

$f(x) = x^2$ for $x < 1$, $2x$ for $x \geq 1$, at $x = 1$

Left-hand limit 1, right-hand limit 2: the values differ.

One side oscillates or is unbounded

That one-sided limit fails to exist, so the two-sided limit fails too.

THE COMMON SLIP

one side Computing only the right-hand limit of the piecewise f gives 2 and the false claim $\lim_{x \to 1} f(x) = 2$.